

What is Data Science?

Data science is an interdisciplinary field that uses **scientific methods, algorithms, and systems** to extract insights and knowledge from data. It combines elements of **statistics, mathematics, programming, machine learning, and domain expertise** to solve real-world problems.

In simple terms, **data science turns raw data into actionable knowledge** that supports decision-making.

Types and Levels of Data

(a) Based on Structure

•Structured Data

- Organized in rows and columns (like in databases, spreadsheets).
- Easy to store, search, and analyze.
- Examples: Sales transactions, student marks, bank records.

•Unstructured Data

- No predefined format or organization.
- Harder to analyze without advanced tools (like NLP, computer vision).
- Examples: Text documents, images, audio, videos, social media posts.

(b) Based on Measurement (Levels of Data)

1.Nominal (Qualitative – Categories without order)

1. Labels or names only, no ranking.
2. Example: Gender (Male/Female), Colors (Red, Blue).

2.Ordinal (Qualitative – Ordered categories)

1. Categories with a meaningful order, but differences aren't measurable.
2. Example: Ratings (Poor, Good, Excellent), Education level.

3.Interval (Quantitative – Ordered with equal intervals, no true zero)

1. Differences are measurable, but zero doesn't mean "nothing."
2. Example: Temperature in Celsius, IQ scores.

4.Ratio (Quantitative – Ordered, equal intervals, true zero)

1. Has all properties of interval plus a meaningful zero.
2. Example: Height, weight, age, income.

(c) Quantitative vs. Qualitative

- **Quantitative Data (Numerical)**

- Can be measured and expressed in numbers.
- Example: Age = 25, Salary = ₹50,000.

- **Qualitative Data (Categorical)**

- Describes attributes or categories.
- Example: Blood group = "A+", Country = "India".

Overview of the Data Science Life Cycle

A typical **data science workflow** includes the following stages:

Step 1. Problem Definition

- Identify the goal: What problem are we solving?
- Example: Predict student performance, detect fraud, recommend movies.

Step 2. Data Collection

Gather raw data from multiple sources:

- Databases (SQL, Excel sheets)
- APIs, web scraping
- IoT sensors, logs
- Surveys, experiments

Step 3. Data Cleaning & Preparation

- Handle **missing values, outliers, inconsistencies**.
- Convert data into usable formats.
- Feature engineering (create new useful variables).

Step 4. Exploratory Data Analysis (EDA)

- Visualizations (histograms, boxplots, scatterplots).
- Statistical summaries (mean, median, correlations).
- Detecting patterns and anomalies.

Step 5. Modeling & Machine Learning

- Choose appropriate models:
 - Regression, Classification, Clustering, Time series.
- Train & test models using libraries like **scikit-learn, TensorFlow, PyTorch**.

Step 6. Evaluation

Metrics depend on the problem type:

- Accuracy, Precision, Recall, F1 (classification)
- RMSE, MAE (regression).

Step 7. Deployment

Deploy models into real-world systems:

Web apps, mobile apps, dashboards.

Step 8. Monitoring & Feedback

- Monitor accuracy over time.
- Retrain models when data changes (concept drift).

Why Data Science is Important

- Helps businesses make **data-driven decisions**.
- Improves **customer experiences** (recommendation engines, chatbots).
- Enhances **healthcare** (disease prediction, drug discovery).
- Powers **automation & AI** (self-driving cars, voice assistants).

Data Science vs. Business Intelligence

1. Focus / Goal

Business Intelligence (BI):

Focuses on **descriptive analytics** (what happened in the past & what is happening now).

Goal: **Reporting & monitoring** to support decision-making.

Data Science (DS):

Focuses on **predictive & prescriptive analytics** (what will happen & what should be done).

Goal: **Prediction, optimization, automation, insights.**

2. Type of Data Used

BI: Mainly **structured data** from databases, spreadsheets, etc.

DS: Uses both **structured + unstructured data** (text, images, videos, IoT, logs, etc.).

3. Tools & Techniques

BI Tools: SQL, Excel, Tableau, Power BI, etc.

DS Tools: Python, R, TensorFlow, scikit-learn, Hadoop → Machine learning, AI, advanced analytics.

4. Methods

•BI:

- Dashboards, Excels, queries, charts.
- Answers: *“How many sales last month?”*

•DS:

- Machine learning, predictive modeling, deep learning, NLP, data mining.
- Answers: *“Can we predict next month’s sales? What factors drive churn?”*

5. End Users

BI: Managers, executives, business analysts (for decision support).

DS: Data scientists, ML engineers, researchers (for insights & automation).

6. Example Scenarios

BI Example:

Retail company uses BI dashboards to **track daily sales, region-wise revenue, and monthly growth.**

DS Example:

The same company uses data science to **predict which customers are likely to churn and recommend personalized offers.**

Data science tools

1. Data Collection & Storage

- **Databases:**
 - **SQL** (MySQL, PostgreSQL, Oracle, MS SQL Server) → Structured data.
 - **NoSQL** (MongoDB, HBase) → Unstructured/semi-structured data.
- **Big Data Tools:** Hadoop, Spark, Google BigQuery.
- **APIs & Web Scraping:** BeautifulSoup, Scrapy, REST APIs.

2. Data Cleaning & Preparation

- **Python Libraries:** Pandas, NumPy, Dask.
- **R Packages:** dplyr, tidyr, data.table.

3. Data Exploration & Visualization

- **Python:** Matplotlib, Seaborn, Plotly, Bokeh.
- **R:** ggplot2, Shiny.
- **BI Tools (for visualization):** Tableau, Power BI, QlikView, Looker.

4. Machine Learning & Modeling

- **Python:** scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Keras.
- **R:** caret, randomForest, mlr.

5. Deep Learning & AI

- TensorFlow, PyTorch, Keras → Neural networks.
- OpenCV → Computer vision.
- Hugging Face Transformers, spaCy, NLTK → NLP (Natural Language Processing).

6. Deployment & Production

- **Model Serving:** Flask, FastAPI, Django (Python web frameworks).
- **Containerization:** Docker, Kubernetes.
- **Cloud Platforms:** AWS (SageMaker, Redshift), Google Cloud AI, Microsoft Azure ML.

7. Workflow & Version Control

- **Version Control:** Git, GitHub, GitLab.
- **Notebooks:** Jupyter Notebook, Google Colab, R Markdown.
- **MLOps Tools:** MLflow, Kubeflow, Airflow (workflow automation).

8. Collaboration & Communication

- Slack, MS Teams, Jira (for project management).
- Confluence, Notion (documentation).

Data Science Use Cases

1. Business & Marketing

- **Customer Segmentation** → Grouping customers based on behavior for targeted marketing.
- **Recommendation Systems** → Amazon, Netflix, and Spotify suggest products/movies/songs.
- **Churn Prediction** → Identifying customers likely to leave and offering retention strategies.

2. Finance & Banking

- **Fraud Detection** → Detect unusual transactions with ML algorithms.
- **Credit Risk Modeling** → Banks assess loan eligibility and default risks.
- **Algorithmic Trading** → Automated trading strategies based on data patterns.

3. Healthcare & Medicine

- **Disease Prediction & Diagnosis** → AI models detect cancer, diabetes, heart disease.
- **Drug Discovery** → Accelerating new drug development using bioinformatics.
- **Personalized Medicine** → Tailoring treatments based on patient data and genetics.

4. Retail & E-commerce

- **Demand Forecasting** → Predict sales to optimize inventory.
- **Dynamic Pricing** → Airlines, hotels, and e-commerce adjust prices in real time.
- **Customer Sentiment Analysis** → Analyze product reviews and social media comments.

5. Manufacturing & Industry

- **Predictive Maintenance** → Forecast machine failures before they happen.
- **Quality Control** → Image recognition to detect defective products.
- **Supply Chain Optimization** → Reduce costs and improve logistics.

6. Government & Public Sector

- **Smart Cities** → Traffic flow optimization, waste management, and energy savings.
- **Crime Prediction & Prevention** → Using data to identify high-risk areas.
- **Policy Making** → Data-driven insights for healthcare, education, and economy.

7. Transportation & Logistics

- **Route Optimization** → Google Maps, Uber, Ola suggest fastest routes.
- **Self-driving Cars** → Tesla, Waymo use data science for autonomous navigation.
- **Fleet Management** → Optimize delivery networks (e.g., FedEx, DHL).

8. Sports & Entertainment

- **Player Performance Analysis** → Used in cricket, football, and basketball for training.
- **Game Strategies** → Data-driven team planning.
- **Fan Engagement** → Personalized content for audiences.

9. Education

- **Student Performance Prediction** → Early identification of at-risk students.
- **Adaptive Learning** → Platforms like Coursera, Khan Academy recommend courses.
- **Exam Monitoring & Cheating Prevention** → AI proctoring tools.